

Executive decision memo

LUCKY CEMENT | MGT-353 | 5 October 2026

Company-Calibrated Synthetic Dataset for Academic Analysis — illustrative figures, calibrated to Lucky Cement's disclosed scale, not actual company records.

Decision: provisional commitment with weekly release

How many tonnes of Ordinary Portland Cement should Lucky Cement plan to supply to an illustrative Karachi/Sindh dealer territory over the next 12 weeks, and how much dispatch-linked production should it commit now versus keep flexible?

12-week forecast	Scheduled commitment	Flexible base	Separate buffer
22,109 t	17,687 t (80%)	4,422 t (20%)	212 t

Recommendation and evidence

For weeks starting 5 October to 21 December 2026, schedule the commitment across the horizon, release dispatch-linked production weekly, retain the 20% flexible balance and reserve access to 20% upside capacity. Do not manufacture the entire horizon now. The calendar seasonal-trend method has the best common-holdout WAPE (7.71%) and MAPE (8.27%); typical error is 106 tonnes/week. Its lead over the naive baseline is modest, so model sophistication alone is not the justification.

Risk and financial consequence

Tracking Signal is -4.12, already outside the assumed +/-4 band, with average overforecasting of 36 tonnes/week. Review the stockout-driven miss and bias before releasing the plan. A fixed base plan under 20% lower demand leaves 4,422 tonnes excess, tying up approximately PKR 75.17m at assumed cost. A 40% dispatch-capacity loss in the first two weeks leaves 661 tonnes unserved even after the buffer, with PKR 2.97m assumed lost contribution. The buffer separately costs PKR 3.60m.

Drivers, review trigger and assumptions

Training regression associates dealer incentives with +372 tonnes/week, controlling for other included drivers. Price evidence is weak; do not infer a real price elasticity or causal uplift. Reforecast weekly and escalate for >10% deviation in either first two weeks, Tracking Signal outside +/-4, or any announced dispatch interruption. Capacity 2,400 t/week, one-week flexible response, unit cost PKR 17,000/t and lost contribution PKR 4,500/t are synthetic assumptions. All quantities concern the illustrative territory, not company-wide sales.